Project IQ - External Libraries & Tools Inventory

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• *Application: Project IQ - Rasa-based IT Support Chatbot

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Backend Dependencies

Core Framework Stack

■ Rasa (v3.6.20)

• Purpose: Open-source conversational AI framework for building the chatbot

• Usage:

  • NLU (Intent/Entity classification)

  • Core dialogue management

  • Policy-based response generation

• Key Files: domain.yml, nlu.yml, stories.yml, rules.yml, data/

• Port: 5005 (Rasa Core server)

• Dependencies: TensorFlow/PyTorch, Sanic (async web framework)

■ Rasa-SDK (v3.6.2)

• Purpose: SDK for developing custom actions in Python

• Usage:

  • Execute custom business logic (GLPI integration, KB search, etc.)

  • ActionGlpiLogin, ActionCreateGlpiTicket, ActionSearchKnowledgeBase, etc.

• Key Files: actions/actions.py, actions/*.py (all action handlers)

• Port: 5055 (Action Server)

■ Pydantic (v1.10.9)

• Purpose: Data validation and settings management

• Usage: Validate request/response schemas, configuration objects
• Key Files: Config models in actions/config_manager.py

■ Typing-extensions (v4.5.0)
• Purpose: Backport of Python typing features for compatibility
• Usage: Type hints in codebase

■ Packaging (v20.9)
• Purpose: Core utilities for handling package versions
• Usage: Version comparison, dependency management

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Web Framework & API

■ Flask (v3.0.0)
• Purpose: Lightweight Python web framework for Admin API
• Usage:
  • REST API endpoints for admin dashboard
  • Static file serving (HTML, CSS, JS)
  • Configuration management routes
  • Analytics endpoints
• Key Files: actions/admin_api.py
• Port: 80/8181 (Admin API)
• Endpoints: /admin/*, /themes/*, /config/*, /health/*

■ Flask-CORS (v4.0.0)
• Purpose: Cross-Origin Resource Sharing support
• Usage: Allow frontend to call backend from different origin
• Key Files: actions/admin_api.py (CORS initialization)

■■ Werkzeug (v3.0.1)
• Purpose: WSGI utilities and HTTP handling
• Usage: HTTP request/response handling (used by Flask)
• Dependencies: Flask dependency

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Database & ORM

■■ PostgreSQL (via psycopg2-binary v2.9.9)
• Purpose: Production-grade relational database
• Usage:
  • Conversation tracker store (messages, sessions)
  • User feedback and ratings storage
  • Analytics data persistence

- Theme and configuration storage

• Key Files: actions/conversation_db.py, actions/analytics_store.py

• Database: Chatbot (default, configurable)

• Tables: conversations, messages, feedback, themes, chat_apps, etc.

• Port: 5432

■ psycopg2-binary (v2.9.9)

• Purpose: PostgreSQL Python adapter/driver

• Usage: Direct database connection and queries

• Key Files: actions/conversation_db.py

■■ SQLAlchemy (<2.0)

• Purpose: Python SQL toolkit and Object Relational Mapper

• Usage:

- Rasa tracker store ORM mapping
- Database connection pooling
- Query building

• Key Files: endpoints.yml (tracker_store configuration)

• Important: Pinned to <2.0 for Rasa 3.6 compatibility

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Secrets & Configuration

■ Infisical SDK (>=1.0.14)

• Purpose: Enterprise secrets management integration

• Usage:

- Retrieve sensitive credentials from Infisical vault
- Manage API keys, tokens securely
- Support multiple environments (dev/prod)

• Key Files: actions/config_manager.py

• Credentials: GLPI tokens, DB passwords, API keys

• Fallback: Local .env file if Infisical unavailable

■ python-dotenv (v1.0.0)

• Purpose: Load environment variables from .env files

• Usage:

- Load database credentials
- Rasa API configuration
- Local development setup

• Key Files: .env (config file), actions/config_manager.py

• Config Keys:

```
DB_HOST, DB_PORT, DB_NAME, DB_USER, DB_PASSWORD GLPI_API_URL, GLPI_APP_TOKEN, GLPI_USER_TOKEN RASA_API_URL
```

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Machine Learning & Search

■ Sentence-Transformers (v2.3.1)

• Purpose: Generate semantic embeddings for Knowledge Base search

• Usage:

  • Create vector embeddings for KB articles

  • Semantic similarity matching

  • Hybrid search (semantic + keyword matching)

• Model: all-MiniLM-L6-v2 (384-dim embeddings)

• Key Files: actions/kb_cache_manager.py

• Dependencies: PyTorch or TensorFlow (auto-installed)

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System Monitoring & Performance

■ psutil (v7.2.1)

• Purpose: Cross-platform library for retrieving system & process information

• Usage:

  • Monitor CPU usage

  • Track memory consumption

  • Disk space monitoring

  • Process status checks

• Key Files: actions/process_monitor.py

• Admin Panel: System Observability dashboard displays real-time metrics

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Data Processing

■ PyYAML (v6.0.1)

• Purpose: Parse and generate YAML files

• Usage:

  • Read domain.yml, nlu.yml, stories.yml, rules.yml

  • Write configuration files

  • Validate YAML syntax

• Key Files: domain.yml, endpoints.yml, multiple action files

• Critical For: Rasa configuration, story/rule definitions

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HTTP Client

■ Requests (v2.31.0)

• Purpose: HTTP library for making API requests

- Usage:
  - Call GLPI REST API (tickets, KB articles)
  - Communicate with Rasa core/actions
  - Call Infisical API
  - External webhook calls
- Key Files: actions/glpi_service.py, actions/config_manager.py, custom actions
- Common Endpoints:
  - GLPI: http://localhost:8083/apirest.php/*
  - Rasa: http://localhost:5005/*
  - Infisical: https://app.infisical.com/api/v1/*
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Production Server (Optional)

■ Waitress (v2.1.2)

- Purpose: WSGI application server for production deployment
- Usage: Alternative to Flask dev server for production
- When Used: Production environment (optional, not in dev setup)
- Benefits: Better concurrency handling, stability
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Testing (Optional)

■ pytest (v7.4.3)

- Purpose: Python testing framework
- Usage: Unit tests, integration tests
- Test Files: tests/ directory (if present)
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Frontend Dependencies

UI Framework

■■ React (v19.2.0)

- Purpose: JavaScript library for building UI components
- Usage:
  - Admin dashboard components
  - Chat widget UI
  - Studio interface
  - Admin analytics/config panels
- Key Files: client_ui/src/components/*.jsx, client_ui/src/App.jsx

■ React-DOM (v19.2.0)

• Purpose: React rendering for web browsers

• Usage: Render React components to DOM

• Key Files: client_ui/src/main.jsx

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Styling & CSS

■ TailwindCSS (v3.4.17)

• Purpose: Utility-first CSS framework

• Usage:

  • Style admin dashboard

  • Style chat widget

  • Responsive design

  • Theme customization

• Key Files: client_ui/tailwind.config.js, client_ui/src/*.css

• Build: Compiled during Vite build

■ PostCSS (v8.4.35)

• Purpose: Tool for transforming CSS

• Usage:

  • Process Tailwind directives

  • Autoprefixing

• Key Files: client_ui/postcss.config.js

■ Autoprefixer (v10.4.18)

• Purpose: PostCSS plugin to add vendor prefixes

• Usage:

  • CSS compatibility across browsers

  • Automatic prefix generation

• Key Files: Part of PostCSS pipeline

■ Tailwind-Merge (v3.4.0)

• Purpose: Merge Tailwind CSS class lists intelligently

• Usage:

  • Avoid conflicting Tailwind classes

  • Theme override handling

• Key Files: client_ui/src/components/*.jsx (conditional styling)

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Icons & UI Components

■ Lucide-React (v0.561.0)

• Purpose: Beautiful, consistent SVG icon library

• Usage:
  • Admin dashboard icons
  • Chat widget icons (send, attach, etc.)
  • UI iconography
• Icons Used: Send, Paperclip, Minimize, Trash, User, Lock, etc.

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Content & Markdown

■ React-Markdown (v10.1.0)
• Purpose: Render Markdown to React components
• Usage:
  • Display markdown formatted bot responses
  • Knowledge Base article display
  • Rich text formatting
• Key Files: client_ui/src/components/*.jsx

■ Remark-GFM (v4.0.1)
• Purpose: GitHub Flavored Markdown plugin for remark
• Usage:
  • Support tables, strikethrough, etc.
  • Enhanced markdown features
• Dependencies: React-Markdown plugin

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Utilities

■ UUID (v13.0.0)
• Purpose: Generate universally unique identifiers
• Usage:
  • Create unique message IDs
  • Session identifiers
  • Widget instance IDs
• Key Files: client_ui/src/components/ChatWindow.jsx

■■ clsx (v2.1.1)
• Purpose: Utility for constructing className strings
• Usage:
  • Conditional CSS class application
  • Theme-based styling
• Key Files: client_ui/src/components/*.jsx

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Animations

■ Framer-Motion (v12.23.26)

• Purpose: Animation library for React

• Usage:

  • Chat widget animations

  • Message transitions

  • UI element animations

  • Smooth page transitions

• Key Files: client_ui/src/components/*.jsx

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Build Tools

Core Build System

■ Vite (v7.2.4)

• Purpose: Next-generation frontend build tool and dev server

• Usage:

  • Development server (npm run dev)

  • Production bundle creation (npm run build)

  • Widget compilation (npm run build:widget)

  • Hot module replacement (HMR)

• Key Files:

  • client_ui/vite.config.js - Main config

  • client_ui/vite.widget.config.js - Widget-specific config

• Output:

  • client_ui/dist/ - Admin dashboard

  • client_ui/dist-widget/ - Embeddable widget (glpi-chat-widget.js)

■ @vitejs/plugin-react (v5.1.1)

• Purpose: React plugin for Vite

• Usage:

  • JSX transformation

  • Fast refresh during development

• Key Files: client_ui/vite.config.js

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Code Quality

■ ESLint (v9.39.1)

• Purpose: JavaScript linter for code quality

• Usage:
  • Find and fix problems in code
  • Enforce coding standards
  • Style consistency
• Key Files: client_ui/.eslintrc.cjs
• Plugins:
  • @eslint/js
  • eslint-plugin-react-hooks
  • eslint-plugin-react-refresh

■ Globals (v16.5.0)
• Purpose: Global variable definitions for ESLint
• Usage:
  • Configure ESLint aware of browser/Node globals

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Type Safety

■ @types/react (v19.2.5)
• Purpose: TypeScript type definitions for React
• Usage:
  • Type checking in React components
  • IDE autocomplete

■ @types/react-dom (v19.2.3)
• Purpose: TypeScript type definitions for React-DOM
• Usage:
  • Type checking for DOM rendering

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Server for Widget

■ http-server (npm, used via script)
• Purpose: Simple HTTP server for static files
• Usage:
  • Serve the compiled widget (dist-widget/)
  • Allow embed testing on different ports
• Command: npx -y http-server dist-widget -p 8181 --cors -c-1
• Port: 8181 (widget serving)
• Flags: CORS enabled, no cache

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External Systems & Services

GLPI (ITSM)

■ GLPI (External System)
• Purpose: IT Service Management platform
• Integration Method: REST API
• API Endpoint: http://localhost:8083/apirest.php
• Operations:
  • Create/read/update tickets
  • Knowledge base articles
  • User authentication
  • Entity/department management
• Authentication: App Token + User Token
• Key Files: actions/glpi_service.py
• Tokens Stored In: Infisical or .env

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Infisical Vault

■ Infisical (External Secrets Service)
• Purpose: Enterprise secrets vault
• Integration Method: HTTP API + SDK
• API Endpoint: https://app.infisical.com/api/v1/
• Operations:
  • Retrieve API tokens
  • Store database credentials
  • Environment-based secrets
• Fallback: Local .env if Infisical unavailable
• Key Files: actions/config_manager.py
• Credentials: Universal Auth client ID & secret

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Rasa Services

■ Rasa Core (Internal Service)
• Purpose: NLU and dialogue management engine
• Port: 5005
• API Endpoints:
  • /webhooks/rest/webhook - Chat messages
  • /model/parse - NLU analysis

- /domain - Domain retrieval
- /conversations/{id}/tracker - Conversation history

■ Rasa Action Server (Internal Service)

• Purpose: Execute custom actions

• Port: 5055

• API Endpoints:

- /webhook - Receive action requests
- /health - Health check

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Infrastructure & DevOps

Containerization

■ Docker (Container Runtime)

• Purpose: Package application in containers

• Key Files: Dockerfile (if present), docker-compose.yml, docker-compose.infisical.yml

• Containers:

- Rasa Core container
- Action Server container
- Admin API container
- PostgreSQL container
- GLPI container
- Nginx container

■ Docker Compose (Container Orchestration)

• Purpose: Define and run multi-container applications

• Configs:

- docker-compose.yml - Base configuration
- docker-compose.infisical.yml - With Infisical integration

• Services: Rasa, Actions, Admin, Database, Nginx, GLPI

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Reverse Proxy & Load Balancing

■ Nginx (Reverse Proxy)

• Purpose: HTTP reverse proxy and load balancer

• Usage:

- Unified entry point (port 80/443)
- Route requests to backend services
- SSL/TLS termination
- Static file serving

• Routes:
  • / → Admin API (port 80)
  • /webhooks/* → Rasa Core (port 5005)
  • /actions/* → Action Server (port 5055)
• Key Files: Nginx config (in deployment)

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Version Control

■ Git (Version Control System)
• Purpose: Source code version control
• Usage:
  • Track code changes
  • Branching and merging
  • Collaborative development
• Key Files: .git/, .gitignore

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Runtime

■ Python 3.x (Runtime)
• Purpose: Python interpreter for backend
• Version: 3.8+ (recommended 3.10+)
• Package Manager: pip
• Virtual Environment: venv/conda

■ Node.js (JavaScript Runtime)
• Purpose: JavaScript runtime for frontend build & serving
• Package Manager: npm
• Key Files: package.json, package-lock.json

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Development Tools & Scripts

Shell Scripts (Windows)

■ Batch/PowerShell Scripts
• start_services.bat - Start all services
• stop_services.bat - Stop all services
• restart_services.bat - Restart all services
• restart_admin_api.bat - Restart admin API
• setup.bat - Initial setup

• backup_database.bat - Backup PostgreSQL

• restore_database.bat - Restore from backup

• health_check.bat - Health check script

■ PowerShell Scripts

• start_all.ps1 - Start services (PS version)

• test_departments.ps1 - Department testing

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Summary Table

Library	Version	Type	Purpose	Key File(s)
---	---	---	---	---
rasa	3.6.20	Framework	NLU + Dialogue	domain.yml, nlu.yml, stories.yml
rasa-sdk	3.6.2	SDK	Custom Actions	actions/actions.py
flask	3.0.0	Framework	Admin API	actions/admin_api.py
flask-cors	4.0.0	Middleware	CORS Support	actions/admin_api.py
psycopg2-binary	2.9.9	Driver	PostgreSQL	actions/conversation_db.py
sqlalchemy	<2.0	ORM	Database Access	endpoints.yml
pydantic	1.10.9	Validation	Data Validation	config models
infisicalsdk	>=1.0.14	SDK	Secrets	actions/config_manager.py
python-dotenv	1.0.0	Config	.env Loading	.env
sentence-transformers	2.3.1	ML	KB Embeddings	actions/kb_cache_manager.py
psutil	7.2.1	Monitoring	System Stats	actions/process_monitor.py
pyyaml	6.0.1	Parser	YAML Processing	All config files
requests	2.31.0	HTTP	API Calls	actions/glpi_service.py
waitress	2.1.2	Server	WSGI Server	(optional)
pytest	7.4.3	Testing	Unit Tests	tests/
react	19.2.0	Framework	UI Components	client_ui/src/
react-dom	19.2.0	Library	DOM Rendering	client_ui/src/
vite	7.2.4	Build Tool	Build System	client_ui/vite.config.js
tailwindcss	3.4.17	CSS	Styling	client_ui/tailwind.config.js
framer-motion	12.23.26	Animation	UI Animation	client_ui/src/components/
lucide-react	0.561.0	Icons	Icon Library	client_ui/src/components/
uuid	13.0.0	Utility	ID Generation	client_ui/src/components/
react-markdown	10.1.0	Component	Markdown Rendering	client_ui/src/components/

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Dependency Graph

```
Project-IQ ■ ■■■ Backend Services ■ ■■■ Rasa Core (v3.6.20) ■ ■ ■■■ TensorFlow/PyTorch ■ ■ ■■■ Sanic
(async web) ■ ■ ■■■ YAML (v6.0.1) ■ ■ ■ ■■■ Action Server (rasa-sdk v3.6.2) ■ ■ ■■■ Flask (v3.0.0) ■ ■
■■■ Requests (v2.31.0) ■ ■ ■■■ psycopg2 (v2.9.9) ■ ■ ■■■ SQLAlchemy (<2.0) ■ ■ ■■■ sentence-transformers
(v2.3.1) ■ ■ ■■■ psutil (v7.2.1) ■ ■ ■■■ Infisical SDK (>=1.0.14) ■ ■ ■■■ python-dotenv (v1.0.0) ■ ■ ■
■■■ Admin API (Flask v3.0.0) ■ ■■■ Flask-CORS (v4.0.0) ■ ■■■ Werkzeug (v3.0.1) ■ ■■■ Static Files
(HTML/CSS/JS) ■ ■■■ Frontend Stack ■ ■■■ Build System: Vite (v7.2.4) ■ ■ ■■■ @vitejs/plugin-react (v5.1.1)
■ ■ ■■■ TailwindCSS (v3.4.17) ■ ■ ■■■ PostCSS (v8.4.35) ■ ■ ■■■ Autoprefixer (v10.4.18) ■ ■ ■ ■■■ UI
Framework: React (v19.2.0) ■ ■ ■■■ React-DOM (v19.2.0) ■ ■ ■■■ Framer-Motion (v12.23.26) ■ ■ ■■■
Lucide-React (v0.561.0) ■ ■ ■■■ UUID (v13.0.0) ■ ■ ■■■ clsx (v2.1.1) ■ ■ ■■■ Tailwind-Merge (v3.4.0) ■ ■
■■■ React-Markdown (v10.1.0) ■ ■ ■■■ Remark-GFM (v4.0.1) ■ ■ ■ ■■■ Code Quality: ESLint (v9.39.1) ■ ■■■
@eslint/js ■ ■■■ eslint-plugin-react-hooks ■ ■■■ eslint-plugin-react-refresh ■ ■■■ External Services ■ ■■■
GLPI (REST API) ■ ■■■ Infisical (Secrets Vault) ■ ■■■ PostgreSQL (Database) ■ ■■■ Nginx (Reverse Proxy) ■
■■■ DevOps ■■■ Docker (Containerization) ■■■ Docker Compose (Orchestration) ■■■ Git (Version Control)
```

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Licensing Summary

Dependency	License	Type
----------	---------	------
Rasa	Apache 2.0	Open Source
Flask	BSD-3-Clause	Open Source
React	MIT	Open Source
TailwindCSS	MIT	Open Source
PostgreSQL	PostgreSQL	Open Source
Docker	Apache 2.0	Open Source
Nginx	BSD-2-Clause	Open Source
sentence-transformers	Apache 2.0	Open Source
Framer-Motion	MIT	Open Source

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Installation Commands

Backend Dependencies

```
pip install -r requirements.txt
```

Frontend Dependencies

```
cd client_ui npm install
```

Build Frontend

```
npm run build npm run build:widget
```

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Environment Requirements

Component	Requirement
----------	----------
OS	Windows, Linux, macOS
Python	3.8+ (recommend 3.10+)
Node.js	16+ (recommend 18+)
RAM	4GB minimum (8GB+ recommended)
Disk	5GB minimum
Database	PostgreSQL 12+
Docker	v20+ (for containerized deployment)

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Critical Version Constraints

■■ Important Pinning Notes

1. SQLAlchemy <2.0
   • Rasa 3.6 is not compatible with SQLAlchemy 2.0+
   • Must be pinned to v1.x series
2. Rasa 3.6.20
   • Requires specific versions of dependencies
   • TensorFlow 2.12+ or PyTorch 1.12+
   • Compatible with Python 3.8-3.11
3. React 19.2.0
   • Modern React with concurrent features
   • Requires Node 16+
4. TailwindCSS 3.4.17
   • Requires PostCSS 8+
   • Node 14+

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Monitoring & Health Checks

Health Check Endpoints

Service	Endpoint	Port
---------	----------	------
Rasa Core	`http://localhost:5005/health`	5005
Action Server	`http://localhost:5055/health`	5055

Admin API	`http://localhost:80/health` or `http://localhost:8181/health`	80/8181
PostgreSQL	Connection via `psycopg2`	5432

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Troubleshooting

Common Dependency Issues

Issue	Cause	Solution
---	---	---
Rasa import error	TensorFlow not installed	`pip install tensorflow`
Flask port conflict	Port 80 already in use	Change port or stop conflicting service
React build fails	Node version too old	Upgrade to Node 16+
DB connection error	PostgreSQL not running	Start PostgreSQL service
CORS errors	Flask-CORS not configured	Check `admin_api.py` CORS setup
Sentence-transformers slow	Model not cached	First run downloads 500MB+ model

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Additional Resources

• [Rasa Documentation](https://rasa.com/docs)
• [Flask Documentation](https://flask.palletsprojects.com)
• [React Documentation](https://react.dev)
• [TailwindCSS Documentation](https://tailwindcss.com)
• [PostgreSQL Documentation](https://www.postgresql.org/docs)
• [Docker Documentation](https://docs.docker.com)

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